

Kit Maker — User Guide

Browse, build and sequence sample drumkits — random kits, full-control kits, and two built-in pattern sequencers.

Kit Maker, by TK & Flurmechanik

The five tabs

Tab	What it's for
Browser	Organise sample packs & kits, audition, analyse and tag, send them onward.
Explosion	Point at a folder, get a finished random kit.
Builder	Define pools & slots yourself, batch-export variations.
Step	16-lane step sequencer driving the kit's RS5K rack.
Euclid	Euclidean pattern generator per kit lane.

A typical flow runs left to right across the tabs: **Browser** to collect and audition your samples → **Explosion** or **Builder** to render kits into RS5K racks → **Step** or **Euclid** to program patterns on those racks. The **Kit Manager** button (top bar) opens a 4×4 rack window that ties it all together. **Esc** or the red dot (top-right) closes the window; your **Theme** choice is remembered.

The installed version is shown next to the **Kit Maker** title. **Right-click that title** for this manual and for the two folders Kit Maker uses — they are not the same place: the script is installed under `Scripts`, while presets and the tag cache are written next to REAPER's own data.

1. Install & requirements

Install through **ReaPack** — the easiest route and it keeps the script up to date:

1. In REAPER: **Extensions** → **ReaPack** → **Import repositories...** and paste the TK-Scripts repository URL: `https://raw.githubusercontent.com/TouristKiller/TK-Scripts/master/index.xml`
2. **Extensions** → **ReaPack** → **Browse packages...**, find **Kit Maker** (by TK & Flurmechanik) and click **Install**.
3. Run it from the **Actions** list (search for *Kit Maker*) — assign it a toolbar button or shortcut if you like.
4. Make sure the dependencies are installed via ReaPack too: **RealmGui** (required — the window needs it to draw) and **SWS/CF** (recommended: folder pickers, audition, "open folder").

Prefer a manual install? Unzip so the TK Kit Maker folder (with `core/`, `ui/`, `data/`) stays intact under `.../REAPER/Scripts/`, then load `TK_Kit_Maker.lua` via **Actions** → **New action** → **Load ReaScript....**

Good to know — The sequencers install a small JSFX engine on the kit track automatically, so timing stays tight even when the window is busy. If the Step/Euclid page says the engine isn't running, check that the kit track's FX aren't bypassed.

2. Browser

Your library hub — collect folders, audition samples, send them onward.

The Browser keeps two kinds of collections side by side. The **PACKS / KITS** button top-left shows which you are in and switches to the other:

- **Sample packs** — raw folders of samples you browse and pull from (feeds Explosion & Builder).
- **Sample kits** — finished kits; from here you can build a playable **RS5K Drum Rack** in REAPER.

Managing collections (left panel)

- **+ Folder** adds a collection; **Rescan** refreshes its file list. Right-click **+ Folder** to set a **start folder** — once one is set, every browse dialog opens there instead of wherever you last browsed. Clear it from the same menu to hand control back to the recent folders.
- One button toggles between the **List** and cover **Tiles** — it shows which you are in; right-click it for the tile size. Drag the splitter to resize.
- A second button cycles **Catalog → Split → Samples**, picking which panel gets the room on a narrow window; right-click steps back.
- Right-click a collection for the full menu: **Rename**, **Pin**, **Move Up/Down**, **Include subfolders**, **Set / Clear / Open Cover**, **Open Folder** and **Remove Collection**.

Auditioning & using samples (right panel)

- **Filter by filename** narrows the list. **Flat list** drops the folder grouping and shows every sample in one run — useful once a tag filter has cut the collection down to a handful spread over many folders. **Folders** puts the grouping back, with **Expand / Collapse** for the headers.
- Preview with **Audition / Stop**, a **volume** slider, and an **Auto** toggle that plays a sample the moment you click it (double-click always plays).
- Right-click a sample to **Load to RS5K on selected track**, drop it into a specific **rack slot**, or **Show in Explorer**.
- **Drag** a sample onto a REAPER track or a Kit Manager pad to load it into RS5K there.
- **Alt+drag** starts a real file drag instead, the kind the operating system handles — so the sample can be dropped into *any* sampler that accepts files (Kontakt, Battery, whatever you use), or into another program entirely. This one needs the **TK Native Helper** extension, installed separately through ReaPack; everything else works without it.

Sample analysis & tags

Kit Maker can measure what a sample actually *sounds* like and tag it — its frequency band, how sharp its attack is, how long it rings out, and whether it is noise or pitched.

Analysis never starts on its own. Under the filter box sits a progress bar with an **Analyse** button that measures the selected collection:

- It runs in the background while you keep browsing, **pauses entirely while REAPER is recording** and eases off during playback.
- **Stop** at any moment — every sample measured so far is kept, and pressing Analyse again simply carries on with what is left.

- Results are cached per file, so a pack is only ever measured once. The ... menu holds *Re-analyse this collection*, *Re-analyse changed files*, the *Show tags panel* toggle and *Clear analysis cache*.
- When a run finishes the line reports how long each sample took, so you can tell a slow drive or format from a fast one.

Re-analyse changed files compares every sample with the file on disk and only re-measures the ones that were edited or replaced. The sample you click on is checked the same way, so a file you just re-rendered shows its new figures rather than yesterday's.

The sample you click on is always measured immediately, so the **tags panel** below the list fills in as you browse. It shows the labels as chips, the spectrum across the seven bands (SUB → AIR), and where the sample sits on each axis. Hover the panel for the raw numbers behind the labels.

Filtering by tag

Next to the filename box sits **Tags**, which narrows the list by what the samples sound like. Only the tags actually present in the collection are offered, each with a count — a pack of kicks will not offer you AIR, and the counts already take your other choices into account, so a chip showing 12 really does leave 12 when you click it.

Within an axis the tags are **or'd** (SUB *or* LOW); across axes they are **and'ed** (SUB/LOW *and* SHORT). Tag filters are not remembered between sessions — a forgotten filter hiding most of your pack is a bad way to start.

Category and **Source** are the exceptions: both are read off the path rather than the audio, so they work on a pack you have not analysed yet. Category is also the one axis where a sample can hold several labels at once — an open hi-hat is a hi-hat too, so **Hihat** returns the open and closed ones alike while **Open hihat** returns just the open. Its counts therefore add up to more than the number of files.

The seventeen categories are **Kick · Snare · Hihat · Open hihat · Closed hihat · Clap · Rim · Tom · Crash · Ride · Cymbal · Perc · Shaker · 808 · Bass · Vocal · FX**, matched on the usual naming conventions (BD, k1k, OHH, CHH, vox, ...). Anything the rules do not recognise is gathered under **Uncategorised** at the end of the list — worth a look now and then, because those same files will not respond to a Builder slot filter or an Explosion pattern either.

Channels splits the collection into **MONO** and **STEREO**. Useful when you are building for hardware: plenty of samplers and grooveboxes either refuse a stereo file or sum it to mono, and summing is where the phase problems come from. This one needed no new analysis — the channel count has always been measured, since it decides whether the Space axis reports anything at all.

Every other tag needs a measurement, so while one of those is selected the unanalysed samples drop out of the list — the popup says how many.

Source only recognises words that describe material rather than decorate a name — acoustic, foley, 808, LinnDrum, Ludwig — so a kit called *Wooden Field Machine* is not mistaken for a room recording. The filename is read first and wins; if the word was only in a folder name the tag shows as ACOUSTIC? in the tags panel, because that is a guess about the pack rather than about the sample.

Filter here, weight there. This is a real filter: ask for the noise hats and you get the noise hats. The **Character** setting in Explosion and Builder does the opposite on the same axes — it weights the pool instead of narrowing it, so a batch of kits keeps its variety. Searching and generating want different things; see *Character* under Explosion.

What is measured, and how far to trust it — frequency band, transient and decay are direct measurements. Noise-vs-pitched is reliable, the finer DEFINED/CHROMATIC split is a judgement call. Dry/room/hall/wet only appears for **stereo** samples: on mono material there is nothing to tell a room apart from a naturally long tail, so Kit Maker says nothing rather than guessing. Texture (clean/warm/crunchy/filthy) is not measured yet.

The heatmap

The **Heatmap** button next to the filter box swaps the file list for a grid: instead of names, you see where your samples actually sit. **File list** puts the names back.

Two measured axes span the grid and the third becomes a weighting slider underneath; the two dropdowns pick which is which, from Frequency, Decay and Transient. The grid runs **4×4 by default to match the pad layout**; the slider beside the axis pickers takes it up to 8×8 for a finer look.

- The grid shows whatever the list is showing, so the **Tags** filter narrows it too — pick Kick there and the grid is a kick grid.
- Cell brightness is how much material sits there, weighted by the slider. Moving the slider re-shades the grid — it never removes samples from it, and **Focus** sets how sharply that weighting falls off.
- **Click** a cell to hear a sample from that region, **right-click** for the usual load/show options.
- Keep clicking the same cell to **walk through it**: you get its best match for the slider first, then the next, and so on through the whole cell without hearing a repeat. **Shift-click** steps back the other way; both wrap around, so you can always get back to the one you liked. Clicking another cell — or moving the slider — starts a fresh walk. The weighting still decides the order, so what you hear on the first click is unchanged.
- The cell you last played is ringed and reads **12/33** — where you are in that walk, out of how many samples the cell holds. On small tiles only the ring and the plain count fit; hover for the position and the filename.

The axes rescale to your selection. Narrow to kicks and the frequency axis zooms in on the range those kicks actually occupy, so they spread over the whole grid instead of piling into one column. It means the axis ends read "the lowest and highest in *this* selection", not absolute values — which is what makes the grid useful for comparing samples of the same kind.

Make rack

At **4×4** the grid is a pad layout, so **Make rack** turns it into a kit in one press: one sample per cell, pad 1 from the bottom-left cell through pad 16 at the top-right. With frequency across and decay up, that means pad 1 is the darkest and shortest sound and pad 16 the brightest and longest, with the whole range covered in between.

- Choose **Fill the selected rack** (the rack whose track is selected in REAPER) or **Create a new rack**. Filling replaces what a pad holds and never grows the rack; if you want a full sixteen, create a new one.
- **Export as a kit folder...** does the same picking but writes the samples to disk instead — copied and renamed, like any other Kit Maker export.
- Empty cells leave their pad empty — if your pack has nothing in a corner, the kit shows it.
- **Re-roll** keeps the layout and draws different samples: every pad holds on to its corner of the sound space. Press it until the contents suit you.
- The weighting slider applies to the picks too, so "spread the spectrum, but all punchy" is a single action.

A cell as a Builder pool

Right-click a cell → **Add cell as Builder Pool**. The samples in that cell become a pool and the Builder opens. Link a slot to it and that slot draws only from that corner of the sound space — so "this pad gets something short and bright, whatever else the kit does" is a thing you can state once and keep.

This is the only pool that is a fixed list of files rather than a folder: what selects the samples is how they measure, not where they live. So it behaves a little differently, on purpose:

- It has no folder list and no **Scan** — there is nothing to rescan it from.
- The list is stored *inside* a preset instead of being rebuilt on load, unlike every other pool.
- It is a snapshot. Re-analysing or adding samples does not change it; make a new one from the cell if you want the pack's current contents.

The name comes from where the cell sits — *Sub Legato, Air Short, Middle* — and is yours to rename in the Builder.

Show kit

The other direction: **Show kit** marks where the selected rack's pads sit in this pack. Each pad's number appears in the cell its sample falls into.

The axes deliberately stay on the *collection's* scale rather than fitting themselves to the kit — a kit rescaled to its own range fills the grid no matter what, and would hide the very thing you are looking for. Four percussion pads bunched in one cell is why a kit sounds flat; an empty stretch of grid is what it is missing. Click there to hear what would fill the gap.

Only meaningful on the collection the kit draws from. Because the axes are scaled to what is on screen, a rack from somewhere else is measured with the wrong yardstick: its pads pile up against the edges and "clustered" stops meaning anything. Kit Maker says so in that case rather than letting it read as a finding. Pads that had to be pushed in from outside the range are drawn dimmed.

Right-click the button to re-read the rack after changing it. Samples the rack uses that were never analysed are measured on the spot — sixteen files is a moment's work.

Sending a collection onward

- From a **PACKS** collection (or its right-click menu): **Use for Explosion** sets it as the Explosion source, or **Add as Builder Pool** creates a pool from it.
- From a **KITS** collection: **Create Drum Rack (RS5K)** builds a ready-to-play rack in REAPER, mapped from note 36 up. A stitched WAV in the folder is left out — it is the whole kit in one file, not a pad sound.
- **Export ... samples as a kit folder...** (right-click the collection you have open) copies what the file list is showing into a new kit folder, numbered and renamed like any other Kit Maker export. The filters *are* the selection: narrow a 213-sample pack with the tag filter and the filename box until the list holds the twelve you want, then export those twelve. The menu item names the count, so you can see what you are about to write.

It only offers itself on the collection currently open, because that is the one the filters apply to. Notes are handed out from 36 upward; past 92 samples there are none left, so the note is left out of the filenames rather than repeated.

- Moved your samples? Missing folders show a **Relink / Auto relink** button; remembered locations relink future racks automatically.

3. Explosion

The fastest route: point at a folder, hit Detonate, get a kit.

1. **Source folder** — pick a folder of samples (toggle **Include subfolders**); a live count shows how many were found.
2. **Kit layout** — the **slot count** (1–128) and the **start note**; the note range is shown live.
3. **Slot pattern** (*optional*) — what kind of sample lands in each slot.
4. **Naming** (*optional*) — the **kit name** (empty = random), a **start word** with **Generate kit name** to build one around it, and a **name in filename** added to every copied sample.
5. **Export** — the **destination folder**, what gets picked and what comes out (see the table below), then **Detonate**. A result popup reports what was copied, with an **Open folder** button.

Steps 1–4 are about the kit; step 5 is about writing it out — which is why the destination lives there, the same as in the Builder.

Options worth knowing

Control	What it does
Slot pattern	Comma-separated categories or keywords (e.g. Kick, Snare, Hihat, Clap) assigned to slots in order and repeated to fill them all. Empty = any sample; unknown words match as "filename contains". The + button inserts known categories.
Start word + Generate kit name	Type an optional seed word, then Generate kit name to auto-name the kit (leave the seed empty for a fully random name).
Stitched WAV + cues	Also joins every sample into one WAV with an embedded cue point per slice — ready for slicers. WAV sources only; other formats are skipped.
Max sample length (s)	Samples longer than this are never picked (and stay out of the stitched WAV). 0 = no limit.
Seed	The number the random pick starts from. See below.
Character	Leans the pick towards a sound — dark, punchy, short — on the axes measured by the analyser. See below.

Seeds — rebuilding a kit, and sharing one

A computer's randomness is a fixed sequence; the **seed** is only where you start reading it. Start at the same place and you get the same kit.

Every detonate runs on a seed, and the one it used is left in the field afterwards. So the kit you liked and pressed past is never gone — untick **New seed each detonate** and press again to get it back.

More usefully, it travels. The same seed over the same samples builds the same kit on anyone's machine, so a good combination can be shared as a number rather than as a folder of files.

That last part only holds if the samples really are the same, which is what the **pack** code under the seed is for. It counts the files and reads their names. Two people whose codes match will get the same kit from the same seed; if the codes differ, one of you has an extra sample or a renamed one, and the seed will build something else — correctly, but not what was meant.

Some fine print worth knowing:

- Kits in a batch are seeded consecutively, so kit 34 of seed 84213 is simply seed 84246 — one kit, shareable on its own.

- **Generate kit name** stays random on every press; it is a button you press until you like the answer. The name a *detonate* generates does follow the seed, so the same seed gives the same kit under the same name.
- Changing anything else — the slot pattern, the character bias, the max length — changes the kit too. The seed reproduces a roll of the dice, not a recipe.
- The **Builder** has the same thing, in its Batch section. There the seed sits on the kit definition rather than on the dialog, so a **preset carries its seed**: a saved Builder setup and its seed rebuild the same kits together. It reproduces more, too — the pools, the slots and the bias, not only the picks.
- Switch on the **Sources log** and the exported kit carries its own recipe: the log opens with the seed and the pack code. The sample paths further down only mean anything on the machine that made the kit; those two lines are the part that travels.

Character

Explosion and Builder can steer the random pick towards a *kind* of sound instead of picking flat random. Three axes, each with a direction and a strength: **Frequency** (SUB → AIR), **Decay** (SHORT → LEGATO) and **Transient** (SOFT → IMPACT).

The crucial part: this **weights** the pool, it does not filter it. Ask for dark and punchy and a batch of 50 kits still gives you 50 different kits — they just all lean the same way. Nothing is ever ruled out, so you keep the happy accidents.

- **Focus** sets how hard to lean. 0 leaves the pool untouched; 1 all but ignores anything off the mark.
- Samples that have not been analysed count as *average* on every axis: they stay in the running but are not favoured. Analyse the pack in the Browser first to get the full effect — the panel tells you how many are measured.
- In the Builder, individual slots can override the kit-wide setting through the **Char** column ("whole kit dark, but this one hat short and sharp").

4. Builder

Full control: define pools and slots yourself, then batch-export.

Start from **Fresh start** or load a preset. The page then walks the same way Explosion does: **1 Pools → 2 Slots → 3 Export**, all three always on screen.

Presets

The **Presets...** button sits above the steps, with the name of the preset you are in beside it — saving and loading is a file operation rather than a step in building a kit.

- **Save** (type a name) stores the current pools + slots; **Update** overwrites a listed one.
- **Load** restores a preset; **Delete** removes it. Presets also appear on the empty-start screen.

Step 1 — Pools

A **pool** is a named group of one or more sample folders.

- **+ Pool**, then set the **Alias**, **+ Add folder(s)** and toggle **Subfolders**.
- **Mode**: *Repeat* (samples may be reused) or *Use up* (no repeats until the pool is exhausted, then reshuffle).

- **Scan** counts available samples. Missing folders can be **Relinked** in place.

Step 2 — Slots

Each row is one sound in the kit. Build the table with **+ Slot / + 16 slots; Renumber notes** numbers them upward from note 36 (C2).

From pattern... is the quick route: type **Kick, Snare, Hihat, Clap**, set how many slots you want, and every slot comes out with its filter already set — and with the pool whose **alias matches the word**, so pools called **Kick / Snare / Hihat** wire themselves up. A pool named **Snares** still answers to **Snare**. The popup shows which pool each word found before you commit, and it replaces the slots you have.

Presets in that popup holds the same ready-made patterns Explosion offers — *Classic 4, Drum machine 16*, and so on — plus your own. Save a pattern on either page and it turns up on the other.

Column	Meaning
#	Slot number / order.
Pool	Which pool this slot draws from. Orange (no pool) = link one before exporting.
Note / Pad	MIDI note and pad number for the slot.
Filter	Restrict this slot to a category or a custom keyword; a live count shows how many pool samples match (red = none match).
Char	Optional per-slot character override; blank means the slot follows the kit-wide setting under Export.
Lock	Pin one exact file so that slot always uses it instead of a random pick.

Quick layout... (next to the slot buttons) fills a full keyboard in one go: white keys alternate between **pool A / B**, black keys all use **one pool** (e.g. **Kick / Snare / Hi-hat**). It **replaces** the slots you have, and says so with the count before you press it.

Step 3 — Export

Destination comes first — where the kits are written, the same place Explosion keeps it.

Naming — number style, note notation, whether to include the alias, and the separator, with a live **Preview** of the resulting filename.

- **Kit name prefix** (empty = random) and a **Start word + Generate kit name** helper.
- **Kit count** (1–200) to make many variations in one run.
- **Seed** — the number the picks start from, with the pack code underneath. See *Seeds* under Explosion; it works the same here, and a preset saves it.
- Optional logs: **MIDI log**, **Sources log**, **Used-samples log** (avoids repeats across sessions), plus **Stitched WAV + cues** and **Max sample length**. The MIDI log is written in REAPER's note-name format (**36 Kick**), so it can be loaded in the MIDI editor under *File → Note names* — or read as a plain list of what sits on which note.
- A line above the button reports readiness — *"16 slots, all linked to a pool"*, or a warning naming how many have no pool, since those stay empty.
- **Batch export** runs with a progress bar; each kit is its own subfolder of copied, renamed samples.

5. Kit Manager (RS5K rack)

The 4x4 pad window behind the playable racks.

Open it with the **Kit Manager** button in the top bar. It shows the selected kit's RS5K rack as a 4x4 pad grid and is where racks are created and maintained — the Browser's **Create Drum Rack** and both sequencers all target this rack.

- **New Kit** creates a brand-new empty RS5K rack in REAPER.
- Per rack: **Rename kit**, a **Cover** image, **Rack color** and **Rack volume**.
- Per pad: load / replace a sample, set the **MIDI input**, a per-pad **Note-off** setting (also used to stop the sound on mouse release), and pad **auto-name**.
- **Relink** finds pads whose sample moved or whose drive letter changed — pick the new folder once and future racks relink automatically.

Racks name their MIDI notes after the instrument, so the piano roll reads **Kick, Snare, Closed hihat** rather than C2, D2, F#2. Where the naming rules recognise nothing, no name is set and REAPER goes on showing the pad track's own name — the sample filename — so those rows never come up blank.

6. Step sequencer

A 16-lane grid driving kit slots 1–16 on the selected rack.

Select the kit's folder track first (the header will say so if you haven't). Each of the 16 lanes is one kit slot; click cells to place steps. Lanes play through the rack's RS5K instances in real time, and a JSFX engine keeps the timing solid.

Per-step editing

Open a lane (click its name; right-click toggles the editor) and pick a **parameter mode** to draw values across its 16 steps:

Mode	Effect
Velocity	How hard each step hits.
Gate / Length	Note length per step (Gate needs RS5K note-off; hidden for one-shot lanes).
Substeps	Ratchets/rolls — retrigger a step several times.
Pitch / Pan / Volume	Per-step RS5K pitch, pan and level offsets.
Probability	Chance a step actually fires, for variation each loop.
Step-mode mask	A per-loop x / . mask (x = trigger, . = skip) so a step can play on some loop repeats and not others.

Per-lane controls

Control	What it does
Solo / Mute	Focus or silence a lane.

Control	What it does
One-shot / Retrig	Play the whole sample ignoring gate, and choose retrigger behaviour.
Note-off (obey)	Whether the lane respects RS5K note-off (needed for Gate/Length to shorten).
Speed	Lane runs at 1×, 0.5× or 2× the master step rate.
Direction	Forward, reverse and other play orders per lane.
Echo	Built-in repeats with adjustable rate, count and velocity mode/delta.
Copy / Paste / Clear	Move lanes around; auto-name labels lanes from their samples.

Pattern pages, presets & playback

- Four **pattern pages** per kit; **copy / paste** a whole page. They work with or without a preset — without one they are kept on the kit track, so they survive closing the script.
- A global **pattern library**: **New**, **Save**, **Rename** and **Delete** presets, selectable per page. Saving a preset stores all four pages.
- Play in **pattern mode** (loop the page) or **song mode** (chain pages); **Host** follows REAPER's transport and tempo.
- **Export to MIDI**: left-click exports one pattern, right-click exports all four (the Step-Mode cycle); a separate control does pattern vs. song export.

7. Euclid sequencer

Generative Euclidean rhythms per kit lane, on the same rack.

Euclid shares the rack and lanes with the Step page but generates each lane's rhythm mathematically: spread a number of hits (**Pulse**) as evenly as possible across a number of **Steps**, then **Rotate** to shift the accent. Select a lane to reveal its knob row.

Knob	What it does
Steps / Pulse / Rotate	Pattern length, number of hits, and rotation of the pattern.
Speed / Sub / Len	Lane speed (0.5×/1×/2×), ratchet substeps, and active pattern length.
Prob	Per-lane probability that a hit fires.
Vel / Gate	Base velocity and note length for the lane.
Pitch / Vol / Pan	RS5K pitch, level and pan for the lane (needs an RS5K instance on the lane).
Attack / Release	RS5K amp envelope shaping per lane.
Vel / Pitch / Vol randomisers	"Dice" that add controlled randomness with an amount level and a seed; re-roll for a new variation while keeping the pattern.

- Lane buttons mirror the Step page: **Solo**, **Mute**, **One-shot**, **Retrig**, **Note-off**.
- Its own preset library (**New**, **Save**, **Rename**, **Delete**), plus **Host** sync, **Export**, and a lane connection-lines toggle.

8. Good to know

- Exports always **copy** samples — your originals are never modified.
- Explosion, Builder and the Browser all fill the same kit structures and share one export engine, so a kit made one way behaves the same everywhere.
- Both sequencers store their patterns on the kit track itself, so each rack carries its own sequences.
- Presets, logs, the pattern library and the analysis cache live under `.../REAPER/TK_Kit_Maker/` alongside the script's data.
- Analysis is stored per file path, so the same sample is only ever measured once no matter how many collections point at it.

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